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L1F24BSCS0506

Lab: Database

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CREATE DATABASE

```
mysql> CREATE DATABASE IF NOT EXISTS Lab_06;
Query OK, 1 row affected (0.01 sec)

mysql> Use Lab_06;
Database changed
mysql> SELECT DATABASE();
+------------+
| DATABASE() |
+------------+
| Lab_06     |
+------------+
1 row in set (0.00 sec)

mysql> |
```

Q1. Create a table **Patient** with following fields

- PatientID varchar (15)
- Name varchar (15)
- Age int
- Gender varchar(1)
- Address varchar(20)
- Disease varchar(10)
- DoctorID varchar(15)

Constraints.

- **PatientID** in Patient, is primary key.

```
mysql> CREATE TABLE IF NOT EXISTS Patients (
    ->     patient_id VARCHAR(15) NOT NULL,
    ->     patient_name VARCHAR(15) NOT NULL,
    ->     patient_age INT NOT NULL,
    ->     patient_gender VARCHAR(1),
    ->     patient_address VARCHAR(20) NOT NULL,
    ->     patient_disease VARCHAR(10) NOT NULL,
    ->     doctor_id VARCHAR(15),
    ->     PRIMARY KEY (patient_id)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> show tables;
+------------------+
| Tables_in_Lab_06 |
+------------------+
| Patients         |
+------------------+
1 row in set (0.01 sec)

mysql> |
```

Q2. Change the datatype of Gender from varchar(1) to char in **Patient** table.

```
mysql> ALter TABLE Patients
    -> Modify COLUMN patient_gender CHAR(1);
Query OK, 0 rows affected (0.19 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Patients;
+-----------------+-------------+------+-----+---------+-------+
| Field           | Type        | Null | Key | Default | Extra |
+-----------------+-------------+------+-----+---------+-------+
| patient_id      | varchar(15) | NO   | PRI | NULL    |       |
| patient_name    | varchar(15) | NO   |     | NULL    |       |
| patient_age     | int         | NO   |     | NULL    |       |
| patient_gender  | char(1)     | YES  |     | NULL    |       |
| patient_address | varchar(20) | NO   |     | NULL    |       |
| patient_disease | varchar(10) | NO   |     | NULL    |       |
| doctor_id       | varchar(15) | YES  |     | NULL    |       |
+-----------------+-------------+------+-----+---------+-------+
7 rows in set (0.01 sec)
```

Q3. Create a table **Doctor** with following fields

- DoctorID varchar (15)
- Name varchar (15)
- Age int
- Gender char
- Address varchar(20)

Constraints.

- **DoctorID** in Doctor, is primary key.

```
mysql> CREATE TABLE IF NOT EXISTS Doctors (
    ->     doctor_id VARCHAR(15) NOT NULL,
    ->     doctor_name VARCHAR(15) NOT NULL,
    ->     doctor_age INT NOT NULL,
    ->     doctor_gender CHAR(1),
    ->     doctor_address VARCHAR(20) NOT NULL,
    ->     PRIMARY KEY (doctor_id)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> show tables;
+------------------+
| Tables_in_Lab_06 |
+------------------+
| Doctors          |
| Patients         |
+------------------+
2 rows in set (0.00 sec)
```

Q4. Now add column **DoctorWard varchar(15)** in Doctor table.

```
mysql> ALTER TABLE Doctors
    -> ADD COLUMN doctor_ward VARCHAR(15);
Query OK, 0 rows affected (0.13 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Doctors;
+----------------+-------------+------+-----+---------+-------+
| Field          | Type        | Null | Key | Default | Extra |
+----------------+-------------+------+-----+---------+-------+
| doctor_id      | varchar(15) | NO   | PRI | NULL    |       |
| doctor_name    | varchar(15) | NO   |     | NULL    |       |
| doctor_age     | int         | NO   |     | NULL    |       |
| doctor_gender  | char(1)     | YES  |     | NULL    |       |
| doctor_address | varchar(20) | NO   |     | NULL    |       |
| doctor_ward    | varchar(15) | YES  |     | NULL    |       |
+----------------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

Q5. There is no relationship between Patient and Doctor table, so make a relation between these two tables using alter command.

Note DoctorID in Patient, should be foreign key from Doctor table.

```
mysql> ALTER TABLE Patients
    -> ADD CONSTRAINT fk_doctor FOREIGN KEY(doctor_id) REFERENCES Doctors(doctor_id)
    -> ON DELETE CASCADE ON UPDATE CASCADE;
Query OK, 0 rows affected (0.13 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Patients;
+-----------------+-------------+------+-----+---------+-------+
| Field           | Type        | Null | Key | Default | Extra |
+-----------------+-------------+------+-----+---------+-------+
| patient_id      | varchar(15) | NO   | PRI | NULL    |       |
| patient_name    | varchar(15) | NO   |     | NULL    |       |
| patient_age     | int         | NO   |     | NULL    |       |
| patient_gender  | char(1)     | YES  |     | NULL    |       |
| patient_address | varchar(20) | NO   |     | NULL    |       |
| patient_disease | varchar(10) | NO   |     | NULL    |       |
| doctor_id       | varchar(15) | YES  | MUL | NULL    |       |
+-----------------+-------------+------+-----+---------+-------+
7 rows in set (0.01 sec)
```

Q6. Create a table **Lab** with following fields

- LabID varchar (15)
- TestDate date
- TestAmount varchar(20)
- PatientID varchar(15)
- DoctorID varchar(15)

Constraints.

- **LabID** in Lab, is primary key.

```
mysql> CREATE TABLE IF NOT EXISTS Labs (
    ->     lab_id VARCHAR(15) NOT NULL,
    ->     test_date DATE NOT NULL,
    ->     amount VARCHAR(20) NOT NULL,
    ->     patient_id VARCHAR(15) NOT NULL,
    ->     doctor_id VARCHAR(15) NOT NULL,
    ->     PRIMARY KEY (lab_id),
    ->     CONSTRAINT fk_lab_patient FOREIGN KEY (patient_id)
    ->         REFERENCES Patients (patient_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE,
    ->     CONSTRAINT fk_lab_doctor FOREIGN KEY (doctor_id)
    ->         REFERENCES Doctors (doctor_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql> show tables;
+------------------+
| Tables_in_Lab_06 |
+------------------+
| Doctors          |
| Labs             |
| Patients         |
+------------------+
3 rows in set (0.01 sec)
```

Q7. Change name of Amount column in Lab to TestAmount.

```
mysql> ALTER TABLE Labs
    -> RENAME COLUMN amount TO test_amount;
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Labs;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| lab_id      | varchar(15) | NO   | PRI | NULL    |       |
| test_date   | date        | NO   |     | NULL    |       |
| test_amount | varchar(20) | NO   |     | NULL    |       |
| patient_id  | varchar(15) | NO   | MUL | NULL    |       |
| doctor_id   | varchar(15) | NO   | MUL | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

Q8. Change the data type of Amount in Lab from varchar(20) to double.

```
mysql> ALTER TABLE Labs
    -> MODIFY COLUMN test_amount DOUBLE;
Query OK, 0 rows affected (0.20 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Labs;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| lab_id      | varchar(15) | NO   | PRI | NULL    |       |
| test_date   | date        | NO   |     | NULL    |       |
| test_amount | double      | YES  |     | NULL    |       |
| patient_id  | varchar(15) | NO   | MUL | NULL    |       |
| doctor_id   | varchar(15) | NO   | MUL | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

Q9. Add a new column with the name of LabNo varchar (10).

```
mysql> ALTER TABLE Labs
    -> ADD COLUMN lab_no VARCHAR(10) NOT NULL;
Query OK, 0 rows affected (0.23 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Labs;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| lab_id      | varchar(15) | NO   | PRI | NULL    |       |
| test_date   | date        | NO   |     | NULL    |       |
| test_amount | double      | YES  |     | NULL    |       |
| patient_id  | varchar(15) | NO   | MUL | NULL    |       |
| doctor_id   | varchar(15) | NO   | MUL | NULL    |       |
| lab_no      | varchar(10) | NO   |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

Q10. Create a table **HospitalRoom** with following fields

- RoomID varchar (15)
- RoomAllocationDate date
- RoomType varchar(20)
- RoomStatus varchar(5)

Constraints.

- **RoomID** in HospitalRoom, is primary key.

```
mysql> CREATE TABLE IF NOT EXISTS HospitalRoom (
    ->     room_id VARCHAR(15) NOT NULL,
    ->     room_allocation_date DATE NOT NULL,
    ->     room_type VARCHAR(20),
    ->     room_status VARCHAR(5) NOT NULL,
    ->     PRIMARY KEY (room_id)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> show tables;
+------------------+
| Tables_in_Lab_06 |
+------------------+
| Doctors          |
| HospitalRoom     |
| Labs             |
| Patients         |
+------------------+
4 rows in set (0.00 sec)
```

Q11. Create a table **PatientBill** with following fields

- BillID varchar (15)
- BillDate date
- Amount varchar(20)
- PatientID varchar(15)

Constraints.

- **BillID** in PatientBill, is primary key.

```
mysql> CREATE TABLE IF NOT EXISTS PatientBill (
    ->     bill_id VARCHAR(15) NOT NULL,
    ->     bill_date DATE NOT NULL,
    ->     bill_amount VARCHAR(20) NOT NULL, -- should be decimal like DECIMAL (10, 2)
    ->     patient_id VARCHAR(15) NOT NULL,
    ->     PRIMARY KEY (bill_id)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> show tables;
+------------------+
| Tables_in_Lab_06 |
+------------------+
| Doctors          |
| HospitalRoom     |
| Labs             |
| PatientBill      |
| Patients         |
+------------------+
5 rows in set (0.01 sec)
```

Q12. There is no relationship between Patient table and PatientBill table, so make a relation between these two tables using alter command.

Note PatientID in PatientBill, should be foreign key from Patient table.

```
mysql> ALTER TABLE PatientBill
    -> ADD CONSTRAINT fk_patient FOREIGN KEY (patient_id)
    ->         REFERENCES Patients (patient_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE;
Query OK, 0 rows affected (0.15 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc PatientBill;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| bill_id     | varchar(15) | NO   | PRI | NULL    |       |
| bill_date   | date        | NO   |     | NULL    |       |
| bill_amount | varchar(20) | NO   |     | NULL    |       |
| patient_id  | varchar(15) | NO   | MUL | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
4 rows in set (0.01 sec)
```

Q13. Rename HosiptalRoom table to Room.

```
mysql> ALTER TABLE HospitalRoom
    -> RENAME TO Room;
Query OK, 0 rows affected (0.05 sec)

mysql> show tables;
+------------------+
| Tables_in_Lab_06 |
+------------------+
| Doctors          |
| Labs             |
| PatientBill      |
| Patients         |
| Room             |
+------------------+
5 rows in set (0.00 sec)
```

DROP DATABASE

```
mysql> DROP DATABASE Lab_06;
Query OK, 5 rows affected (0.12 sec)

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| Books_shop         |
| Shirts_db          |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| x                  |
+--------------------+
7 rows in set (0.00 sec)
```